

Appendix A

Study 1: Participant Information Sheet

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

You are being invited to take part in this research project. Before you decide to do so, please take your time to read this information sheet carefully. Your participation is entirely voluntary. Thank you for considering participating in this study.

What is the purpose of the study?

In this study, we aim to use “SnapAppy”, a smartphone app as a psychological intervention based on digital photography for informal caregivers. The purpose is to explore the smartphone photography as a positive psychology intervention through capturing the positive events and happy moments, then reviewing and reflecting them.

Why have I been invited?

You have been invited to take part in our study as a caregiver of a person who is living with chronic health conditions, disabilities or frailty due to ageing. We aim to involve at least 30 participants in this study, to use the smartphone-based positive psychology intervention for one month.

Do I have to take part?

It is up to you to decide whether or not to take part. Your participation is entirely voluntary. If you do decide to take part, you would be asked to sign a consent prior to any further procedures. You are free to withdraw from the research study at any time and without giving any reason.

Am I eligible to take part?

In order to take part, you must be a caregiver of a person who is living with chronic health conditions, disabilities or frailty due to ageing. You would not be able to take part if you:

1. Are not a caregiver
2. Do not have access to a smartphone.

3. Are unable to understand English instructions.
4. Are currently receiving any kind of psychological sessions that may affect the overall results.

What are the possible benefits of taking part?

We hope that you will find this research interesting and we hope that your contribution in this study will help us understand the role of the smartphone-based positive psychology intervention for informal caregivers.

What are the possible disadvantages and risks of taking part?

There are no risks involved in this study. Nevertheless, you can withdraw at any point if you feel uncomfortable.

What will happen to me if I take part?

You will receive a recorded video, where we will explain the functionality of SnapAppy and a short tutorial on how to use it. Then, you have to log in to the app, which involves three options: Facebook, Google, or other email address. Immediately after logging in the app, you will be presented with some screens explaining the required steps that you should follow to use the app. If needed, you can access the “help” button within the menu at any time during the study. If you have any questions or need clarification, you can contact the researchers, and we can conduct a briefing session via video conferencing facilities such as Zoom to address your queries.

Then, you are expected to use the SnapAppy app for at least 30 days for the study; however, you will be encouraged to engage with the app beyond this period if you have the desire. The data beyond the study period will be stored in the secured cloud, but will not be used for analysis without further consent.

You will also be filling in daily mood reports and common psychological surveys every 10 days, including the Positive and Negative Affect Schedule (PANAS) and the Satisfaction with Life Scale (SWLS).

You will be encouraged to take photos of positive moments, events, and experiences using the SnapAppy app on your smartphone and you will be able to review those photos anytime you want. Moreover, you will have the option to write a description for each photo about the positive moment, event, or experience and its effect on your feelings. You will be encouraged to freely express your thoughts, emotions, and feelings associated with the experienced moment or event. In addition, you will receive a notification to log your mood at the end of each day. you will be asked “How was your mood today, for the whole day in general?” and could respond on a 7-point Likert scale with clearly marked extremes: -3 (Bad) to +3 (Good).

Will my taking part in this project be kept confidential?

All information obtained in this study will be kept strictly confidential. The collected data including any quotations from the participants which are published will be anonymised. We will not include your

name, and we will remove any other potentially identifying information about you. The researcher will keep data for less than 12 months before destroying all data.

What will happen to the results of the study?

The app reports back to the researcher a list of activities (i.e. photo taking and annotation, photo timeline and review, psychological surveys, and retention and user engagement) and responses (i.e. daily mood reports, PANAS, and SWLS) and also the photos. Photos will be stored in the secured cloud, not on the phone. When the study is completed, we would analyse the data we collect and report the findings. It would be reported in an appropriate scientific journal or presented at a conference. You will not be identified in any report or publication. If you wish to receive a copy of any reports resulting from the research, please ask us to include you in our mailing list.

Contact Details:

If you have any enquiries, please feel free to contact

Researcher: Raya Al Habsi

Address:

Email:

Mobile:

Appendix B

Study 1: Informed Consent Form

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

Name of researcher: Raya Al Habsi

Participants Identification Number:

1. Taking part in the study

I confirm I have read and understand the information for the above study. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.

Yes : No

I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason.

Yes : No

2. Use of the information in the study

I understand that my responses will be anonymised before analysis. I give permission for members of the research team to have access to my anonymised responses I understand that the data could be used for publication.

Yes : No

I understand that my responses will only be used for research purposes.

Yes : No

I agree to take part in the above research project.

Yes : No

Name of participant:

Signature:

Date:

Name of the researcher:

Signature:

Date:

To be signed and dated in presence of the participant

Appendix C

Study 1: Demographic Questionnaire

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

SnappAppy Username:

a. Age group:

1. 18-24
2. 25-34
3. 35-44
4. 45-54
5. 55-64
6. 65 above

b. Gender:

1. Male
2. Female

c. Education:

1. Early years
2. Primary
3. Secondary

4. Further Education (FE)
5. Higher Education (HE)
6. Others (please specify)

d. Employment status:

1. Employed Full-Time (40 or more hours/week)
2. Employed Part-Time (less than 40 hours/week)
3. Freelancer
4. Not in Labour Force (retired, homemaker, etc.)
5. Unemployed
6. Others (please specify)

e. Marital status:

1. Single
2. In a relationship
3. Married
4. Separated
5. Divorced
6. Widowed

f. Relationship (with reference to care recipient):

1. Spouse
2. Parent
3. Son/Daughter
4. Sibling
5. Others (please specify)

Appendix D

Study 1: Authentic Happiness Inventory(AHI)

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

Instruction: Please read each group of statements carefully. Then pick the one statement in each group that best describes the way you have been feeling for the past week, including today. Be sure to read all of the statements in each group before making your choice next to the statements.

1. Group 1:

- I feel like a failure.
- I do not feel like a winner.
- I feel like I have succeeded more than most people.
- As I look back on my life, all I see are victories.
- I feel I am extraordinarily successful.

2. Group 2:

- I am usually in a bad mood.
- I am usually in a neutral mood.
- I am usually in a good mood.
- I am usually in a great mood.
- I am usually in an unbelievably great mood.

3. Group 3:

- When I am working, I pay more attention to what is going on around me than to what I am doing.

- When I am working, I pay as much attention to what is going on around me as to what I am doing.
- When I am working, I pay more attention to what I am doing than to what is going on around me.
- When I am working, I rarely notice what is going on around me.
- When I am working, I pay so much attention to what I am doing that the outside world practically ceases to exist.

4. Group 4:

- My life does not have any purpose or meaning.
- I do not know the purpose or meaning of my life.
- I have a hint about my purpose in life.
- I have a pretty good idea about the purpose or meaning of my life.
- I have a very clear idea about the purpose or meaning of my life.

5. Group 5:

- I rarely get what I want.
- Sometimes, I get what I want, and sometimes not.
- Somewhat more often than not, I get what I want.
- I usually get what I want.
- I always get what I want.

6. Group 6:

- I have sorrow in my life.
- I have neither sorrow nor joy in my life.
- I have more joy than sorrow in my life.
- I have much more joy than sorrow in my life.
- My life is filled with joy.

7. Group 7:

- Most of the time I feel bored.
- Most of the time I feel neither bored nor interested in what I am doing.
- Most of the time I feel interested in what I am doing.

- Most of the time I feel quite interested in what I am doing.
- Most of the time I feel fascinated by what I am doing.

8. Group 8:

- I feel cut off from other people.
- I feel neither close to nor cut off from other people.
- I feel close to friends and family members.
- I feel close to most people, even if I do not know them well.
- I feel close to everyone in the world.

9. Group 9:

- By objective standards, I do poorly.
- By objective standards, I do neither well nor poorly.
- By objective standards, I do rather well.
- By objective standards, I do quite well.
- By objective standards, I do amazingly well.

10. Group 10:

- I am ashamed of myself.
- I am not ashamed of myself.
- I am proud of myself
- I am very proud of myself.
- I am extraordinarily proud of myself.

11. Group 11:

- Time passes slowly during most of the things that I do.
- Time passes quickly during some of the things that I do and slowly for other things.
- Time passes quickly during most of the things that I do.
- Time passes quickly during all of the things that I do.
- Time passes so quickly during all of the things that I do that I do not even notice it.

12. Group 12:

- In the grand scheme of things, my existence may hurt the world.

- My existence neither helps nor hurts the world.
- My existence has a small but positive effect on the world.
- My existence makes the world a better place.
- My existence has a lasting, large, and positive impact on the world.
-

13. Group 13:

- I do not do most things very well.
- I do okay at most things I am doing.
- I do well at some things I am doing.
- I do well at most things I am doing.
- I do really well at whatever I am doing.

14. Group 14:

- I have little or no enthusiasm.
- My enthusiasm level is neither high nor low.
- I have a good amount of enthusiasm.
- I feel enthusiastic doing almost everything.
- I have so much enthusiasm that I feel I can do most anything.
-

15. Group 15:

- I do not like my work (paid or unpaid).
- I feel neutral about my work.
- For the most part, I like my work.
- I really like my work.
- I truly love my work.

16. Group 16:

- I am pessimistic about the future.
- I am neither optimistic nor pessimistic about the future.
- I feel somewhat optimistic about the future.
- I feel quite optimistic about the future.

- I feel extraordinarily optimistic about the future.

17. Group 17:

- I have accomplished little in life.
- I have accomplished no more in life than most people.
- I have accomplished somewhat more in life than most people.
- I have accomplished more in life than most people.
- I have accomplished a great deal more in my life than most people.

18. Group 18:

- I am unhappy with myself.
- I am neither happy nor unhappy with myself—I am neutral.
- I am happy with myself.
- I am very happy with myself.
- I could not be any happier with myself.

19. Group 19:

- My skills are never challenged by the situations I encounter.
- My skills are occasionally challenged by the situations I encounter.
- My skills are sometimes challenged by the situations I encounter.
- My skills are often challenged by the situations I encounter.
- My skills are always challenged by the situations I encounter.

20. Group 20:

- I spend all of my time doing things that are unimportant.
- I spend a lot of time doing things that are neither important nor unimportant.
- I spend some of my time every day doing things that are important.
- I spend most of my time every day doing things that are important.
- I spend practically every moment every day doing things that are important.

21. Group 21:

- If I were keeping score in life, I would be behind.
- If I were keeping score in life, I would be about even.

- If I were keeping score in life, I would be somewhat ahead.
- If I were keeping score in life, I would be ahead.
- If I were keeping score in life, I would be far ahead.

22. Group 22:

- I experience more pain than pleasure.
- I experience pain and pleasure in equal measure.
- I experience more pleasure than pain.
- I experience much more pleasure than pain.
- My life is filled with pleasure.

23. Group 23:

- I do not enjoy my daily routine.
- I feel neutral about my daily routine.
- I like my daily routine, but I am happy to get away from it.
- I like my daily routine so much that I rarely take breaks from it.
- I like my daily routine so much that I almost never take breaks from it.

24. Group 24:

- My life is a bad one.
- My life is an OK one.
- My life is a good one.
- My life is a very good one.
- My life is a wonderful one.

Appendix E

Study 1: The Orientation to Happiness Questionnaire(OTH)

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

Instruction: Each item required a respondent to answer on a 5-point scale the degree to which the item applied (“1 = very much unlike me” through “5 = very much like me”).

1. My life serves a higher purpose.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

2. In choosing what to do, I always take into account whether it will benefit other people.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

3. I have a responsibility to make the world a better place.

- Very much unlike me

- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

4. My life has a lasting meaning.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

5. What I do matters to society.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

6. I have spent a lot of time thinking about what life means and how I fit into its big picture.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

7. Life is too short to postpone the pleasures it can provide.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

8. I go out of my way to feel euphoric.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

9. In choosing what to do, I always take into account whether it will be pleasurable.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

10. I agree with this statement: "Life is short – eat dessert first."

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

11. I love to do things that excite my senses.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

12. For me, the good life is the pleasurable life.

- Very much unlike me
- Somewhat unlike me
- Neutral

- Somewhat like me
- Very much like me

13. Regardless of what I am doing, time passes very quickly.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

14. I seek out situations that challenge my skills and abilities.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

15. Whether at work or play, I am usually “in a zone” and not conscious of myself.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

16. I am always very absorbed in what I do.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

17. In choosing what to do, I always take into account whether I can lose myself in it.

- Very much unlike me

- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

18. I am rarely distracted by what is going on around me.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

Appendix F

Study 1: Positive Relationships and Accomplishment-Scale (PRA)

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

Instruction: Each item required a respondent to answer on a 5-point scale the degree to which the item applied (“1 = very much unlike me” through “5 = very much like me”).

1. When choosing what to do, I always take into account if I can do it together with others.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

2. A good life means to me that I can share it with others.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

3. What really matters in life is to be on good terms with other people.

- Very much unlike me

- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

4. Real happiness (roaring with laughter, celebrating success, feeling joy, etc.) can only be experienced with other people.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

5. Other people are the best and the most reliable remedy against the adversities in life.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

6. There are things in my life that I absolutely want to achieve.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

7. There is nothing better than having achieved a goal I was aspiring to.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

8. I have ambitions.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

9. Most things I do give me a feeling of accomplishment.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

10. I am able to complete most things I do successfully.

- Very much unlike me
- Somewhat unlike me
- Neutral
- Somewhat like me
- Very much like me

Appendix G

Study 1: The Positive and Negative Affect Schedule (PANAS)

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

Instruction: This scale consists of a number of words that describe different feelings and emotions. Read each item below and then indicate to what extent you have felt this way over the past 10 days.

1. Interested

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

2. Distressed

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

3. Excited

1. Very slightly or not at all

2. A little
3. Moderately
4. Quite a bit
5. Extremely

4. Upset

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

5. Strong

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

6. Guilty

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

7. Scared

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

8. Hostile

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

9. Enthusiastic

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

10. Proud

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

11. Irritable

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

12. Alert

1. Very slightly or not at all
2. A little
3. Moderately

4. Quite a bit

5. Extremely

13. Ashamed

1. Very slightly or not at all

2. A little

3. Moderately

4. Quite a bit

5. Extremely

14. Inspired

1. Very slightly or not at all

2. A little

3. Moderately

4. Quite a bit

5. Extremely

15. Nervous

1. Very slightly or not at all

2. A little

3. Moderately

4. Quite a bit

5. Extremely

16. Determined

1. Very slightly or not at all

2. A little

3. Moderately

4. Quite a bit

5. Extremely

17. Attentive

1. Very slightly or not at all

2. A little
3. Moderately
4. Quite a bit
5. Extremely

18. Jittery

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

19. Active

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

20. Afraid

1. Very slightly or not at all
2. A little
3. Moderately
4. Quite a bit
5. Extremely

Appendix H

Study 1: Satisfaction With Life Scale (SWLS)

Exploring the Use of Smartphone-Based Positive Psychology Intervention in Reducing Burden and Emotional Distress for Informal Caregivers

Instruction: Below are five statements that you may agree or disagree with. Using the 1 - 7 scale below, indicate your agreement with each item by choosing the appropriate number.

1. In most ways my life is close to my ideal.

- 7 - Strongly agree
- 6 - Agree
- 5 - Slightly agree
- 4 - Neither agree nor disagree
- 3 - Slightly disagree
- 2 - Disagree
- 1 - Strongly disagree

2. The conditions of my life are excellent.

- 7 - Strongly agree
- 6 - Agree
- 5 - Slightly agree
- 4 - Neither agree nor disagree
- 3 - Slightly disagree
- 2 - Disagree
- 1 - Strongly disagree

3. I am satisfied with my life.

- 7 - Strongly agree
- 6 - Agree
- 5 - Slightly agree
- 4 - Neither agree nor disagree
- 3 - Slightly disagree
- 2 - Disagree
- 1 - Strongly disagree

4. So far I have gotten the important things I want in life.

- 7 - Strongly agree
- 6 - Agree
- 5 - Slightly agree
- 4 - Neither agree nor disagree
- 3 - Slightly disagree
- 2 - Disagree
- 1 - Strongly disagree

5. If I could live my life over, I would change almost nothing.

- 7 - Strongly agree
- 6 - Agree
- 5 - Slightly agree
- 4 - Neither agree nor disagree
- 3 - Slightly disagree
- 2 - Disagree
- 1 - Strongly disagree